OP-TEE Shared Secure Peripherals

Part of the TACOS project

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Table of contents

Preface **4**

I Setting up & building OP-TEE **5**

1 Prerequisites **6**
1.1 Ubuntu . . . 6
1.1.1 Changes from original 2 . . . 7
1.2 Google AOSP repo . . . 7
1.3 QEMU . . . 8

2 Building & running OP-TEE **9**
2.1 QEMU . . . 9
2.1.1 Download of repositories . . . 9
2.1.2 Toolchains . . . 9
2.1.3 Build OP-TEE . . . 9
2.1.4 Running OP-TEE . . . 10
2.2 Sabre Lite . . . 10
2.2.1 Download of repositories . . . 10
2.2.2 Output folder for buildroot . . . 11
2.2.3 Build OP-TEE, Linux and the i.MX6 BSP . . . 11
2.2.4 Creating a bootable SD Card . . . 12
2.2.5 Connecting the BD-SL-i.MX6 Serial . . . 12

3 Developing changes for OPTEE-OS **13**
3.1 QEMU . . . 13
3.2 SABRE Lite . . . 13

4 Debugging OP-TEE **15**
4.1 QEMU . . . 15
4.1.1 Download and install gdb for ARM . . . 15
4.1.2 Gdb settings . . . 15
4.1.3 Debugging OP-TEE . . . 16
4.2 SABRE Lite . . . 18

5 Custom Pseudo Trusted Application **19**
5.1 Example PTA . . . 19

6 Serial interrupt in secure world **27**
6.1 Interrupt handler . . . 27
6.2 Serial console driver . . . 28
6.2.1 Handling serial input . . . 28
6.3 CPU masks . . . 29

II Design & implementation of a driver for Shared Secure Peripherals **30**

7 Shared Secure Peripherals for OP-TEE **31**
7.1 Normal world callback on secure world interrupt . . . 31
7.1.1 Feature state . . . 32
7.1.2 Using asynchronous notifications . . . 32
7.1.3 Practical overview . . . 33
7.2 SSP driver using callbacks . . . 35

8 Linux kernel Modules (aka device drivers) **39**
8.1 Built-in modules . . . 39
8.2 Loadable kernel modules . . . 41

9 Implementation into OP-TEE OS **42**
9.1 Build instructions . . . 42
9.1.1 QEMU . . . 42
9.1.2 SABRE Lite . . . 43
9.2 Technical details . . . 44

References **48**

Preface

This is the documentation of the TACOS project.

Part I

Setting up & building OP-TEE

1 Prerequisites

- Ubuntu 20.04 (only this distribution tested)
- Google AOSP repo
- QEMU

1.1 Ubuntu

This guide is based on a clean Ubuntu 20.04 installation in a VM. It should be possible to repeat the results on any Linux distribution and any hypervisor platform, as long as the required packages are available.

At least 60GiB free disk space is required. If the disk is not large enough and the build process is started, the disk will fill, possibly leading to system crashes.

The first thing to do when the operating system is installed, is to update the package manager for the right architectures. Run the following in a command line:

```
sudo dpkg --add-architecture i386
sudo apt update
```

Then the required packages can be installed:

```
sudo apt install android-tools-adb android-tools-fastboot autoconf \
        automake bc bison build-essential ccache cscope curl
 ↪  device-tree-compiler \
        expect flex ftp-upload gdisk iasl libattr1-dev libcap-dev \
        libfdt-dev libftdi-dev libglib2.0-dev libgmp-dev libhidapi-dev \
        libmpc-dev libncurses5-dev libncursesw5-dev libpixman-1-dev
 ↪  libpython2.7 \
        libssl-dev libtool make mtools netcat ninja-build python-crypto \
```

```
        python3-crypto python-is-python3 python-pyelftools python3-pycryptodome
↪  \
       python3-pyelftools python3-serial rsync unzip uuid-dev xdg-utils xterm \
        xz-utils zlib1g-dev
```

1.1.1 Changes from original 2

- python-serial is not available for Ubuntu 20.04. This package has thus been removed from the install procedure. This should not have any impact, because only python3 is used.
- As python3 is the only python version available on Ubuntu 20.04, a package is added that links `python` to `python3`. This might be troublesome on systems where Python 2 is installed and existing tools use Python 2 by calling `python`. If this is the case, change these tools to use `python2` explicitly.

1.2 Google AOSP repo

The repo tool from the Google Android Open Source Project (AOSP) allows you to manage multiple git repositories using manifest files. OP-TEE uses these manifests to combine all projects into a single development environment based on the specific architecture to be used.

Obviously, because repo uses git, git needs to be installed and setup on the system. Configure git with your email and name.

```
sudo apt install git
git config --global user.email "<email>"
git config --global user.name "<name>"
```

After that, repo can be installed:

```
mkdir -p ~/bin
PATH="${HOME}/bin:${PATH}"
curl https://storage.googleapis.com/git-repo-downloads/repo > ~/bin/repo
chmod a+rx ~/bin/repo
```

Because repo is installed in `~/bin`, this directory needs to be added to the path. This should already been taken care of by Ubuntu, this can be checked if the following is present in `~/.profile`. If not, it should be added to the end of the file.

```
 set path so it includes user's private bin if it exists
if [ -d "$HOME/bin" ] ; then
    PATH="$HOME/bin:$PATH"
fi
```

1.3 QEMU

Next, QEMU should be installed. This can be done by executing

```
sudo apt install qemu
```

2 Building & running OP-TEE

2.1 QEMU

2.1.1 Download of repositories

OP-TEE uses a central repository containing the manifest files for repo for all architectures. To download all necessary git repositories of the OP-TEE project for the QEMUv7 architecture, execute the following commands:

```
mkdir -p <project-dir>
cd <project-dir>
repo init -u https://github.com/Distrinet-TACOS/manifest.git -m tacos-qemu.xml
repo sync -j4 --no-clone-bundle
```

2.1.2 Toolchains

The toolchains required to build any part of OP-TEE are included in one of the repositories downloaded in the previous step. To download the actual toolchains, execute

```
cd <project-dir>/build
make -j2 toolchains
```

2.1.3 Build OP-TEE

The OP-TEE project has three main parts: the operating system, the client API and the trusted applications. These three parts can be built on their own, but to get started it is easier to build the whole system at once. This is as simple as executing

```
cd <project-dir>/build
make -j `nproc`
```

This process will take some time.

2.1.4 Running OP-TEE

QEMU and OP-TEE can be launched by executing

```
cd <project-dir>/build
make run -j `nproc`

# After the command prompt (qemu) appears, press c to start execution.
c
```

This will launch QEMU in the current terminal and two new terminals, one serial connection to the secure world and one to the normal world. In the normal world terminal, you can login using the users `root` or `test`, after which it is possible to run programs in the normal world that interact with the secure world.

To test if everything went according to plan, run `xtest` once logged in. If there are no obvious errors, the build process should have completed successfully. You can run the example programs by executing any of the `optee_example_*` programs present in the path.

The command `make run-only -j `nproc`` is also available to run QEMU without building the project.

2.2 Sabre Lite

2.2.1 Download of repositories

The i.MX6 build system is based on a customized Buildroot. The TACOS related changes are maintained in our fork of Buildroot. To initialize the build environment, create a project directory and clone the buildroot into this directory.

```
mkdir -p <project-dir>
cd <project-dir>
git clone https://github.com/Distrinet-TACOS/buildroot.git
```

i.MX6 specific layer on buildroot called buildroot-external-boundary is provided by boundary devices. TACOS related modifications on this repo are maintained in a forked repository at buildroot-external-boundary repository. Check out this repository in the same project folder as mentioned above.

```
cd <project-dir>
git clone https://github.com/Distrinet-TACOS/buildroot-external-boundary.git
```

2.2.2 Output folder for buildroot

Use the following command to overlay the i.MX6 related configurations, the Board Support Package (BSP), OPTEE configurations and TACOS related configurations on the Buildroot setup. The following command will create a directory named output and place the configurations in this folder.

```
cd <project-dir>
make BR2_EXTERNAL=$PWD/buildroot-external-boundary/ -C buildroot/
 ↪  O=$PWD/output nitrogen6x_optee_defconfig
```

2.2.3 Build OP-TEE, Linux and the i.MX6 BSP

The simplest way to bootstrap is to issue a make command. This will fetch all the components and build Linux, OPTEE and OPTEE applications. The Linux image is bounded to the OPTEE image and an SD card image is created.

```
cd <project-dir>/output
make BR2_JLEVEL="$(nproc)" all
```

This process will take some time.

2.2.4 Creating a bootable SD Card

The previous process will create an SD card image in the directory `<project-dir>/output/images/sdcard.img`. You can use an SD card flash tool such as balenaEtcher to flash the image on an SD Card.

2.2.5 Connecting the BD-SL-i.MX6 Serial

The BD-SL-i.MX6 board has two UART connections. The UART connection labled 'Console' will be connected to OPTEE while the other UART will be connected to the Linux running in the Normal world. Once the UART cables are connected and their interfaces are attached to separate console windows. Power up the BD-SL-i.MX6 board and you will see boot logs for OPTEE on the UART labeled console and the Linux logs on the other UART. You can now log in to linux and look around or start an example client application available onboard.

It might be that only one serial connection is used to output both the normal and the secure world consoles. To change this, during the first moments of the boot process, press any key to stop the autoboot. Enter the following in the terminal that appears.

```
setenv console ttymxc0
saveenv
reboot
```

The device should now use both serial connections.

3 Developing changes for OPTEE-OS

3.1 QEMU

It is very easy to make changes to the OPTEE-OS components on QEMU. Just change what you want in the respective directory in `<project-dir>` and rebuild in `<project-dir>/build`.

3.2 SABRE Lite

Because Buildroot is meant as more of a deployment than a development system, making changes to the components requires some work. We will be following the Buildroot documentation.

1. First, clone the repositories of the components you want to in a separate directory. Let's use `<changes-dir>` for this example.
2. Create a file in `<project-dir>/output` called `local.mk` and add the following lines for each component that you cloned locally (in this case OPTEE-OS and Linux):

   ```
   OPTEE_OS_OVERRIDE_SRCDIR = <changes-dir>/optee_os
   LINUX_OVERRIDE_SRCDIR = <changes-dir>/linux
   ```

   This will tell Buildroot that it should not pull the build files from the remote repositories, but should instead use the locally cloned files.
3. Then issue the following command after every change you made to these files (again for OPTEE-OS and Linux):

   ```
   make BR2_JLEVEL="$(nproc)" optee-os-rebuild linux-rebuild all
   ```

   This will copy the files from the local clones to `<project-dir>/output/build/...`

> **Note**
>
> It might be necessary to first run
>
> ```
> make optee-os-dirclean linux-dirclean
> ```
>
> to remove the current build files. This *will* remove all files in `<project-dir>/output/build/...`, so make sure that any changes are backed up first.

4 Debugging OP-TEE

To debug OP-TEE, some additional steps need to be taken to prepare everything.

4.1 QEMU

4.1.1 Download and install gdb for ARM

Normally, if this guide was followed and the toolchains were built accordingly, gdb should already be present in `<project-dir>/toolchains`. The necessary tools can be added to the path by adding the following in `.profile`.

```
if [ -d "<project-dir>/toolchains/aarch32/bin" ] && [ -d
 ↪  "<project-dir>/toolchains/aarch64/bin" ]; then
        PATH="<project-dir>/toolchains/aarch32/bin:<project-
         ↪  dir>/toolchains/aarch64/bin:$PATH"
fi
```

4.1.2 Gdb settings

To make the initialization of gdb easier, add the following at the end of `~/.gdbinit`:

```
set print pretty on

define optee
    handle SIGTRAP noprint nostop pass
    symbol-file <project-dir>/optee_os/out/arm/core/tee.elf
    target remote localhost:1234
end
```

```
document optee
    Loads and setup the binary (tee.elf) for OP-TEE and also connects to the
 ↪  QEMU
    remote.
end
```

4.1.3 Debugging OP-TEE

Because OP-TEE uses Address Space Layout Randomization (ASLR), it is necessary to disable it when starting QEMU. This is done by adding the flag `CFG_CORE_ASLR=n`.

Compilation error

If OP-TEE was compiled before, the compilation process can generate an error similar to:

```
-- The C compiler identification is unknown
-- Check for working C compiler:
 ↪  <project-dir>/out-br/host/bin/arm-linux-gnueabihf-gcc
-- Check for working C compiler:
 ↪  <project-dir>/out-br/host/bin/arm-linux-gnueabihf-gcc -- broken
CMake Error at /usr/share/cmake-3.16/Modules/CMakeTestCCompiler.cmake:60
 ↪  (message):
  The C compiler

    "<project-dir>/out-br/host/bin/arm-linux-gnueabihf-gcc"

  is not able to compile a simple test program.

  It fails with the following output:
  ...
```

This problem originates from mixing compilations with and without `gdbserver` activated. The solution is to completely delete `<project-dir>/out-br` and to start the compilation process again.

Open two terminals. One will be the terminal where QEMU will be running, the other will be used to debug using gdb. Like before, execute the following with the added flag, but **do not press c**!

```
cd <project-dir>/build
make run CFG_CORE_ASLR=n -j `nproc`
```

Once QEMU is started, start gdb in the second terminal:

```
cd <project-dir>/toolchains/aarch32/bin
./arm-linux-gnueabihf-gdb -q
```

When the gdb prompt appears, run `optee` to load the symbols and to connect to QEMU. The output below should appear.

```
optee
> SIGTRAP is used by the debugger.
> Are you sure you want to change it? (y or n) [answered Y; input not from
 ↪  terminal]
> warning: No executable has been specified and target does not support
> determining executable automatically.  Try using the "file" command.
> 0x00000000 in ?? ()
```

Then it is possible to refer to the symbols in the OP-TEE p. To demonstrate, let's set a breakpoint on `tee_entry_std(...)`, an often called function and continue execution. Th

```
Set breakpoint
b tee_entry_std
> Breakpoint 1 at 0xe1347ce: file core/tee/entry_std.c, line 528.

 Resume execution. Breakpoint should be hit.
c
> Continuing.
> [Switching to Thread 1.2]
>
> Thread 2 hit Breakpoint 1, tee_entry_std (arg=0xe300000, num_params=2) at
 ↪  core/tee/entry_std.c:528
> 528             return __tee_entry_std(arg, num_params);
```

Now it is possible to further explore OP-TEE using gdb.

4.2 SABRE Lite

Debugging Linux on hardware boards is pretty difficult. `printk` statements can be used to print execution flow and memory values to the console.

5 Custom Pseudo Trusted Application

Trusted Applications (TA) normally run in user mode in the Trusted Execution Environment. They are dynamically loaded when executed. When you want to run them in kernel mode, a Pseudo TA (PTA) is necessary. During compilation, PTA's are statically linked against the OP-TEE kernel.

It is possible to create a PTA by adding the sources to `optee_os/core/pta` and including them in the Makefile in the same directory. This way the PTA is included in the os when it is built. It is then possible to create an application for the normal world that calls this PTA using its UUID and the right parameters, just like a normal TA.

The example host application is created by copying an existing example application from `optee_examples` and removing the `ta` subfolder (as there will be no TA compiled separately). The Makefile of the host application should be updated with the right binary name and the build instructions for a TA should be removed from the parent (C)Makefiles.

5.1 Example PTA

This example shows a simple PTA that prints some text when called from the normal world.

```
// File: optee_os/core/pta/testpta.c

#include <kernel/pseudo_ta.h>

#define PTA_NAME "test.pta"
#define PTA_TEST_PRINT_UUID { 0xd4be3f91, 0xc4e1, 0x436c, \
    { 0xb2, 0x92, 0xbf, 0xf5, 0x3e, 0x43, 0x04, 0xd5 } }

#define PTA_TEST_PRINT 0

static TEE_Result test_print(uint32_t type, TEE_Param p[TEE_NUM_PARAMS]) {
```

```
    IMSG("This is a test of a pseudo trusted application.");

    return TEE_SUCCESS;
}

/*
 * Trusted Application Etnry Points
 */

static TEE_Result invoke_command(void *session __unused, uint32_t cmd,
                                 uint32_t ptypes,
                                 TEE_Param params[TEE_NUM_PARAMS]) {
  switch (cmd) {
    case PTA_TEST_PRINT:
        return test_print(ptypes, params);
        break;
    default:
        break;
  }

  return TEE_ERROR_NOT_IMPLEMENTED;
}

pseudo_ta_register(.uuid = PTA_TEST_PRINT_UUID, .name = PTA_NAME,
    .flags = PTA_DEFAULT_FLAGS, .invoke_command_entry_point = invoke_command);
```

```
# Added to optee_os/core/pta/sub.mk

ifeq ($(PLATFORM),vexpress-qemu_virt)
srcs-y += testpta.c
endif
```

The code for the host application:

```
// File: host/main.c

/*
```

```
 * Copyright (c) 2016, Linaro Limited
 * All rights reserved.
 *
 * Redistribution and use in source and binary forms, with or without
 * modification, are permitted provided that the following conditions are met:
 *
 * 1. Redistributions of source code must retain the above copyright notice,
 * this list of conditions and the following disclaimer.
 *
 * 2. Redistributions in binary form must reproduce the above copyright notice,
 * this list of conditions and the following disclaimer in the documentation
 * and/or other materials provided with the distribution.
 *
 * THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS"
 * AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE
 * IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE
 * ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE
 * LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR
 * CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF
 * SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS
 * INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN
 * CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE)
 * ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE
 * POSSIBILITY OF SUCH DAMAGE.
 */

#include <err.h>
#include <stdio.h>
#include <string.h>

/* OP-TEE TEE client API (built by optee_client) */
#include <tee_client_api.h>

#define PTA_TEST_PRINT_UUID { 0xd4be3f91, 0xc4e1, 0x436c, \
    { 0xb2, 0x92, 0xbf, 0xf5, 0x3e, 0x43, 0x04, 0xd5 } }

#define PTA_TEST_PRINT 0
```

```c
int main(void)
{
    TEEC_Result res;
    TEEC_Context ctx;
    TEEC_Session sess;
    TEEC_Operation op;
    TEEC_UUID uuid = PTA_TEST_PRINT_UUID;
    uint32_t err_origin;

    /* Initialize a context connecting us to the TEE */
    res = TEEC_InitializeContext(NULL, &ctx);
    if (res != TEEC_SUCCESS)
        errx(1, "TEEC_InitializeContext failed with code 0x%x", res);

    /*
     * Open a session to the "hello world" TA, the TA will print "hello
     * world!" in the log when the session is created.
     */
    res = TEEC_OpenSession(&ctx, &sess, &uuid,
                   TEEC_LOGIN_PUBLIC, NULL, NULL, &err_origin);
    if (res != TEEC_SUCCESS)
        errx(1, "TEEC_Opensession failed with code 0x%x origin 0x%x",
            res, err_origin);

    /*
     * Execute a function in the TA by invoking it, in this case
     * we're incrementing a number.
     *
     * The value of command ID part and how the parameters are
     * interpreted is part of the interface provided by the TA.
     */

    /* Clear the TEEC_Operation struct */
    memset(&op, 0, sizeof(op));

    /*
     * Prepare the argument. Pass a value in the first parameter,
     * the remaining three parameters are unused.
```

```
     */
    op.paramTypes = TEEC_PARAM_TYPES(TEEC_NONE, TEEC_NONE,
                     TEEC_NONE, TEEC_NONE);

    /*
     * TA_HELLO_WORLD_CMD_INC_VALUE is the actual function in the TA to be
     * called.
     */
    printf("Invoking PTA to print test.");
    res = TEEC_InvokeCommand(&sess, PTA_TEST_PRINT, &op, &err_origin);
    if (res != TEEC_SUCCESS)
        errx(1, "TEEC_InvokeCommand failed with code 0x%x origin 0x%x",
            res, err_origin);
    printf("Test printed.");

    /*
     * We're done with the TA, close the session and
     * destroy the context.
     *
     * The TA will print "Goodbye!" in the log when the
     * session is closed.
     */

    TEEC_CloseSession(&sess);

    TEEC_FinalizeContext(&ctx);

    return 0;
}
```

The Makefiles of the new host application:

```
# File: Makefile

export V?=0

# If _HOST or _TA specific compilers are not specified, then use CROSS_COMPILE
HOST_CROSS_COMPILE ?= $(CROSS_COMPILE)
```

```
.PHONY: all
all:
    $(MAKE) -C host CROSS_COMPILE="$(HOST_CROSS_COMPILE)"
     ↪  --no-builtin-variables

.PHONY: clean
clean:
    $(MAKE) -C host clean
```

```
# File: host/Makefile

CC      ?= $(CROSS_COMPILE)gcc
LD      ?= $(CROSS_COMPILE)ld
AR      ?= $(CROSS_COMPILE)ar
NM      ?= $(CROSS_COMPILE)nm
OBJCOPY ?= $(CROSS_COMPILE)objcopy
OBJDUMP ?= $(CROSS_COMPILE)objdump
READELF ?= $(CROSS_COMPILE)readelf

OBJS = main.o

CFLAGS += -Wall -I../ta/include -I$(TEEC_EXPORT)/include -I./include
#Add/link other required libraries here
LDADD += -lteec -L$(TEEC_EXPORT)/lib

BINARY = optee_example_testpta

.PHONY: all
all: $(BINARY)

$(BINARY): $(OBJS)
    $(CC) $(LDFLAGS) -o $@ $< $(LDADD)

.PHONY: clean
clean:
    rm -f $(OBJS) $(BINARY)
```

```
%.o: %.c
    $(CC) $(CFLAGS) -c $< -o $@
```

```
# File: CMakeLists.txt

project (optee_example_testpta C)

set (SRC host/main.c)

add_executable (${PROJECT_NAME} ${SRC})

target_include_directories(${PROJECT_NAME}
            #    PRIVATE ta/include
               PRIVATE include)

target_link_libraries (${PROJECT_NAME} PRIVATE teec)

install (TARGETS ${PROJECT_NAME} DESTINATION ${CMAKE_INSTALL_BINDIR})
```

```
# File: Android.mk

###################### optee-testpta ######################
LOCAL_PATH := $(call my-dir)

include $(CLEAR_VARS)
LOCAL_CFLAGS += -DANDROID_BUILD
LOCAL_CFLAGS += -Wall

LOCAL_SRC_FILES += host/main.c

# LOCAL_C_INCLUDES := $(LOCAL_PATH)/ta/include

LOCAL_SHARED_LIBRARIES := libteec
LOCAL_MODULE := optee_example_testpta
LOCAL_VENDOR_MODULE := true
LOCAL_MODULE_TAGS := optional
include $(BUILD_EXECUTABLE)
```

```
# include $(LOCAL_PATH)/ta/Android.mk
```

6 Serial interrupt in secure world

To accept input on the serial console of the secure world, it is necessary to register an interrupt on that serial console. To do this, two things need to be done: a interrupt handler needs to be defined and the handler must be registered for the interrupt. The console interface might be different for QEMU or Sabre Lite, but the general way of working should be the same.

6.1 Interrupt handler

The interrupt handler is defined according to the `itr_handler` type in `<interrupt.h>`. This struct needs to be initialized with the interrupt location `it` (in this case `IT_CONSOLE_UART`), the flags for the interrupt (`ITRF_TRIGGER_LEVEL`) and the handler function. This function can accept a parameter that is the `itr_handler` structure. When defining this struct, it is (probably) necessary to explicitly state that the interrupt handling is put in the right part of the binary. This is done by declaring `DECLARE_KEEP_PAGER(console_itr);`.

```
struct itr_handler {
    size_t it;
    uint32_t flags;
    itr_handler_t handler;
    void *data;
    SLIST_ENTRY(itr_handler) link;
};

static enum itr_return handler_function(struct itr_handler *h)
{
    // Handling code
    return ITRR_HANDLED;
}
```

```
static struct itr_handler handler = {
    .it = IT_CONSOLE_UART,
    .flags = ITRF_TRIGGER_LEVEL,
    .handler = handler_function,
};
DECLARE_KEEP_PAGER(handler);
```

When the interrupt handler is declared and defined, it can be used to enable the interrupt. First, the interrupt handler must be registered and then the interrupt can be enabled. Adding the interrupt handler is done by `itr_add(&handler)` and enabling the interrupt is done by `itr_enable(handler.it)`.

6.2 Serial console driver

To accept input from the serial console during the handling of the interrupt, it is necessary to have access to the console chip and driver. Because the driver is initialized during the boot sequence of the platform, the only way to get this access is by hijacking this process, disabling the standard interrupt enabled by the boot process and call our own function to save the serial console access location.

The original interrupt setup happens in the file `optee_os/core/arch/arm/plat-vexpress/main.c`. A macro called `driver_init(...)` is called with the function that enables the serial interrupt. This macro ensures that the interrupt enabling routine is called during the boot process. To prevent this from happening, the simplest solution is to comment out this line.

To save the access location of the console interface, the function `console_init(void)` in the same file calls a function `register_serial_console(...)` which is a wrapper for an assign operation. We can imitate this function to do the same for us, by defining our own pointer to the serial chip and using this pointer to access the serial console.

```
static struct serial_chip *serial_chip;
```

6.2.1 Handling serial input

The serial console functions as follows: when data is sent to the serial console (for example by pressing a key in the console), it is accepted by the driver, the data is stored in a buffer and the cpu is notified by means of an interrupt. This interrupt invokes our interrupt handler function.

It is in this function that we are able to retrieve the data from the buffer. The interrupt keeps triggering if the buffer is not empty, thus it is necessary to empty the buffer after getting its contents. The following while loop gets a single character from the buffer and prints it as long as there are characters left in the buffer. It is here that we use the reference to the serial console.

```
static enum itr_return handler_function(struct itr_handler *h)
{
    while (serial_chip->ops->have_rx_data(serial_chip)) {
        int ch = serial_chip->ops->getchar(serial_chip);
        DMSG("new cpu %zu: got 0x%x", get_core_pos(), ch);
    }

    return ITRR_HANDLED;
}
```

6.3 CPU masks

Each system may have multiple cpus. When assigning an interrupt handler to a certain interrupt and enabling it, this interrupt is assigned to all cpus. This means that when an interrupt happens, both cpus will handle that interrupt, calling the same handler twice leading to undefined behavior.

To prevent this from happening, one can enable cpu masks on an interrupt to define which cpu is responsible for handling that interrupt. This is done by calling the function `itr_set_affinity(<interrupt>, <cpu_mask>)` with the interrupt from the handler and a cpu mask that is a sequence of bits, where a 1 at position n enables the handling of an interrupt by the cpu with number n.

It is not yet possible to handle interrupts on a cpu selected at interrupt handle time.

Part II

Design & implementation of a driver for Shared Secure Peripherals

7 Shared Secure Peripherals for OP-TEE

Currently, for *Trusted Execution Environments* (TEE), there exist no drivers that can operate as regular devices (think of character devices in Linux operating systems) and enable interaction between user space applications in the normal world and the TEE subsystem in a transparent fashion. Such a driver could take all the technicalities out of the communication with these TEE subsystems. This is useful in the case of writing an adapter between existing applications, which are already interacting with interfaces of regular devices, and the TEE subsystem, such that the applications do not need to be rewritten.

A *Shared Secure Peripheral* (SSP) driver for OP-TEE consists of two parts: a *Normal World SSP Controller* and a *Secure World Controller*. These two controllers work together to form a bridge across the gap between both worlds that allows devices in the normal world to react to interrupts in the secure world. Using such a driver makes it possible for any normal world, user space applications to interoperate with code running in the secure world without needing major refactoring.

7.1 Normal world callback on secure world interrupt

Because of the design of TEEs, it is not possible to directly communicate with the normal world from the secure world. It is thus not immediately possible to react to a secure world interrupt in the normal world.

Say that a secure interrupt is registered on a serial interface of the secure world and that this interrupt event is of importance to the normal world and requires special handling there. The interrupt infrastructure dictates that the secure world handles this interrupt with an earlier defined interrupt handler. The *Secure Monitor* preempts the normal world and starts execution of this handler in the secure world. But how do we notify the normal world of this event?

The normal world is able to communicate with the secure world via well defined channels. This means that threads on the Linux based operating system can easily open communication channels to trusted applications in the secure world. This happens through the well known GlobalPlatform Client API. The normal world can launch calls into the secure world with some kind of argument passing or shared memory,

as described in the Client API. In this way, the secure world is able to populate those memory locations with data of use to the normal world.

It is thus possible to approximate communication from the secure world to the normal world by globally saving data or state of the interrupt handler and its results, which can be polled by the normal world. This is however an inefficient solution, as the normal world needs to do constant polling to the secure world, and defeats the purpose of interrupts.

The solution to this problem is a newly developed mechanism, aptly called *Asynchronous Notifications*. This is a feature that is added to the Linux kernel, but it is still in the middle of the review process and is thus not yet generally available. To check out this feature, you can clone the `tacos-qemu-callback.xml` manifest file as explained earlier in the Download of repositories section. This section will cover the implementation details of this feature.

7.1.1 Feature state

This notification feature is explained in the OP-TEE documentation in the section about asynchronous notifications. It consists of two active RFCs, one on the optee_os repository and on the linaro-swg linux repository. Because the changes to the linux repository are actually changes to the Linux kernel itself, these changes need to be upstreamed to the actual Linux kernel. Although this feature is generally well received by the kernel maintainers, this review process is lengthy, with multiple back and forths between the Linaro developers and the maintainers. It generally takes a few months until the changes are accepted. The changes to optee_os are on hold until the upstreaming process is finished.

7.1.2 Using asynchronous notifications

Luckily, the mechanism that is currently being reviewed is in a working state, which means we can pull it in our working copy and use it to our hearts content. The changes have already been applied to our fork of the linux kernel, and can be pulled from git as explained in the introduction of this section.

7.1.2.1 Secure world to normal world

The notification to the normal world is achieved through triggering a non-secure interrupt. The triggering of this interrupt is platform specific and done via `itr_raise_pi()`. This call is wrapped in a function called `notif_send_async`, which can be called in an interrupt handler in the secure world. An

example implementation can be found in `optee_os/core/arch/arm/plat-vexpress/main.c`. When this non-secure interrupt has been triggered, the Linux kernel in the normal world launches the normal interrupt handling process. At system startup, a handler for this kind of notification interrupt has been registered, so that handler is used.

The *hard* interrupt handler `notif_irq_handler`[1] of a threaded interrupt with the name `optee_notification` is registered during the initialization of the kernel. This handler is responsible for deciding what to do with the interrupt thrown, based on the value of the notification. This value is collected by the function called `get_async_notif_value`. The only value that is currently implemented is `OPTEE_SMC_ASYNC_NOTIF_VALUE_DO_BOTTOM_HALF`, but the implementation could easily be extended to allow for different notifications.

Because the interrupt handler in the Linux kernel is a *threaded interrupt* it is also possible to do further handling in the interrupt handler thread called `notif_irq_thread_fn`. This is the appropriate way to execute more complex interrupt handling that could block the kernel for quite some time.

7.1.2.2 Normal world to secure world

On the secure side, a *notification driver* can be registered, should you want to use the asynchronous notification feature as intended. An example implementation can be found in the file `optee_os/core/arch/arm/plat-vexpress/main.c`. However, to write some kind of driver split over the two worlds, we are not fully using this feature, but just using half of the communication channel: from the secure to the normal world.

7.1.3 Practical overview

Figure 7.1 shows the sequence of events defined by the asynchronous notifications feature. The OP-TEE driver is the default driver that manages the TEE and thus OP-TEE OS. This driver always registers an interrupt handler for the asynchronous notifications interrupt line on start up. The PTA also registers an interrupt handler, this time for the external hardware interrupt line.

When this hardware interrupt is caught by the PTA, the PTA saves the notification value [2] and sends an asynchronous notification to the OP-TEE driver by triggering a non-secure software interrupt. This allows the PTA to finish execution before the OP-TEE driver handles the interrupt.

[1] `/drivers/tee/optee/notif.c`
[2] This value is manually programmed, based on the interrupt the handler will be handling.

When the OP-TEE driver interrupt handler starts handling the software interrupt, it asks for the notification value using a fast call to the OP-TEE OS. Based on this value the driver may choose to do a multitude of things. In the current implementation state, the only value that is implemented is, as explained before, `OPTEE_SMC_ASYNC_NOTIF_VALUE_DO_BOTTOM_HALF`, which launches a yielding call into the secure os. This yielding call is possible because the interrupt handler in the driver is a threaded interrupt handler. This threaded interrupt handler is implemented in such a way that the notification value is requested in the *hard interrupt context*, but the bottom half yielding call is made in the threaded interrupt context. This yielding call can thus be preempted, functionality that is not inherently present in OP-TEE OS.

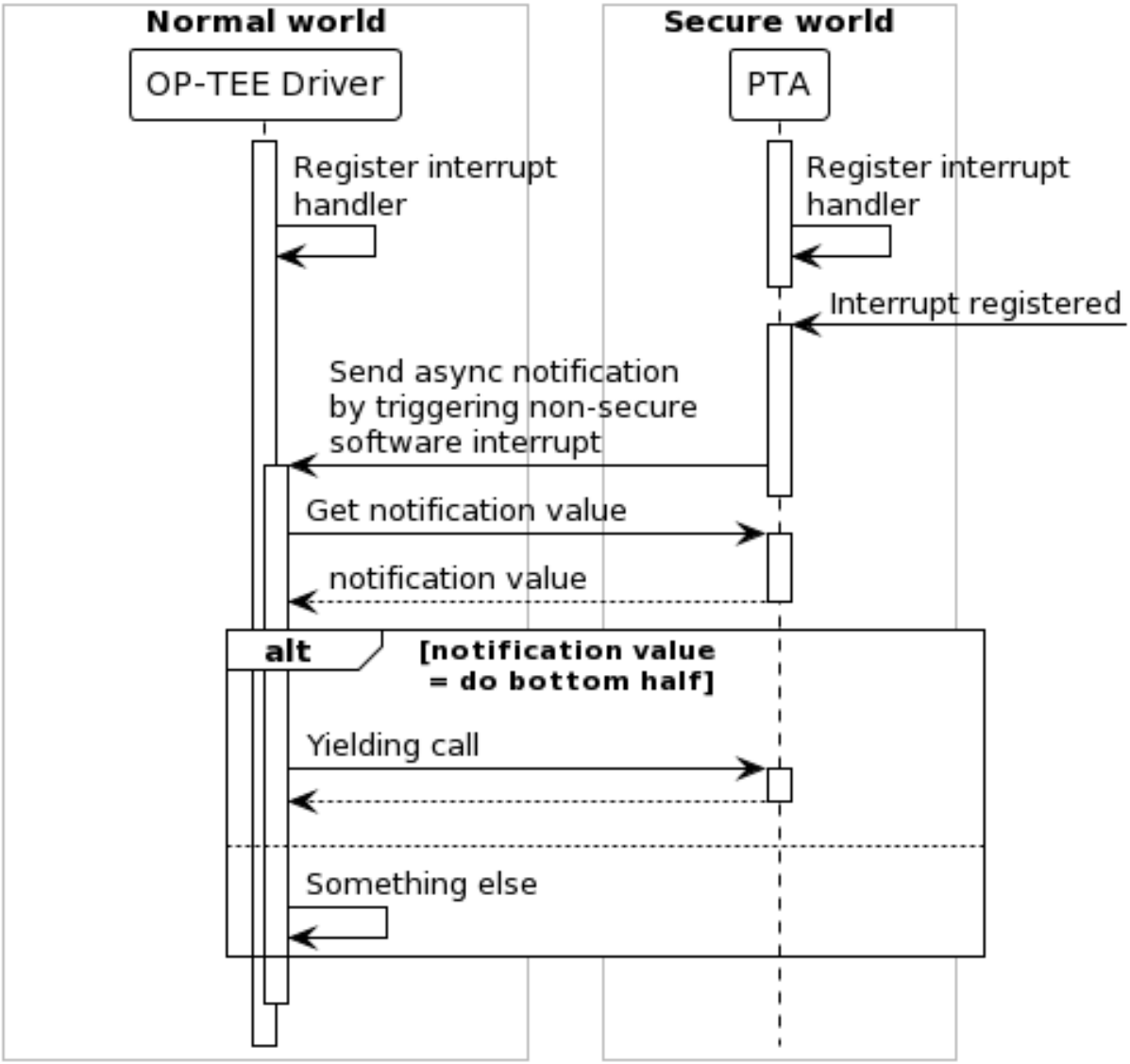


Figure 7.1: Sequence diagram showing the way asynchronous notifications work.

7.2 SSP driver using callbacks

Given the new feature described in the previous section, it is possible to construct a driver that crosses over the normal and the secure world barrier. We created a pair of drivers, a *normal world SSP driver* and

a *secure world SSP driver*, that are responsible for providing access to SSPs. The secure SSP driver directly interacts with the SSP hardware, while the normal SSP driver needs to go through the secure driver for their interaction with that hardware. Figure 7.2 shows the overview of the architecture of the drivers and the system supporting and provided by its functionality.

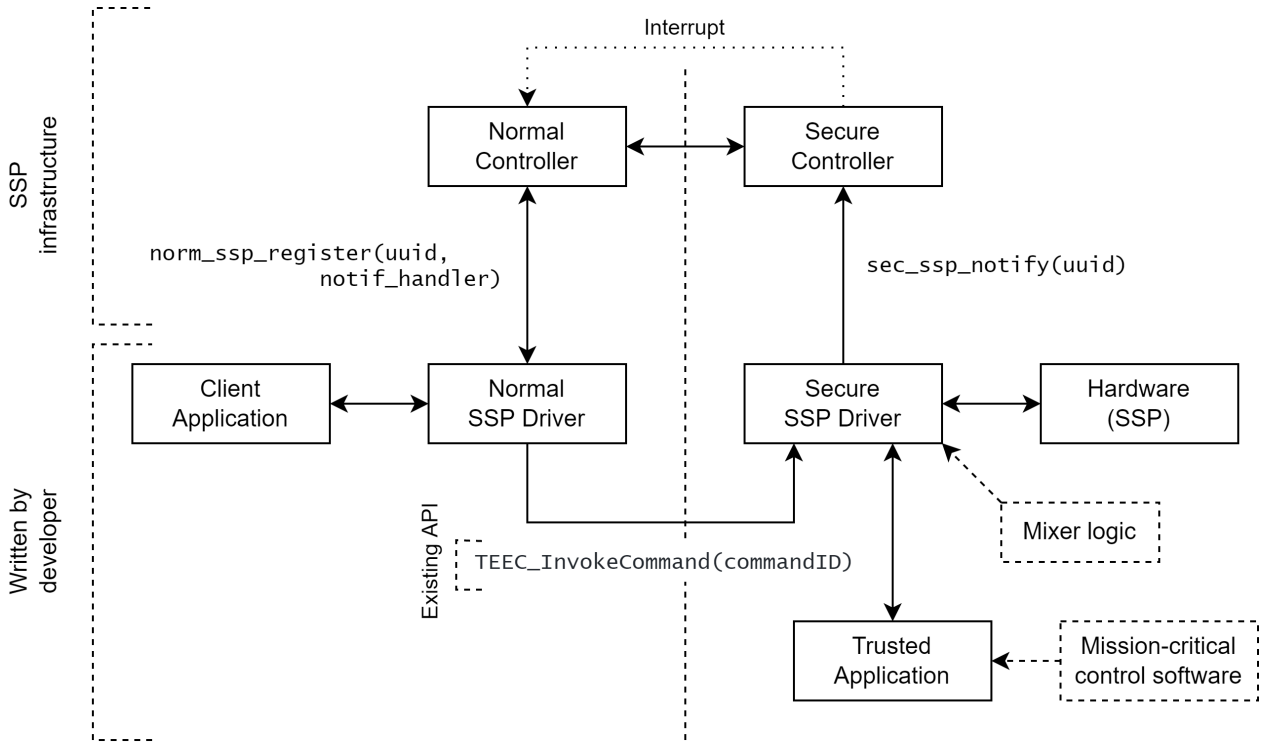


Figure 7.2: Component diagram for the SSP system.

Figure 7.2 also shows that communication from the normal to the secure world is delivered by an existing API (the GlobalPlatform TEE Client API). This API provides access to TAs running in the secure world based on their identifying UUID, and secured by access control. The normal SSP driver may thus call arbitrary functionality, implemented by the developer of a specific application, of the secure SSP driver. The secure SSP driver may thus also implement some kind of mixer logic if required.

> **Note**
>
> *Mixer Logic*
> Consider a display as an SSP. This display normally shows output of a normal world application, but overlays some output of a secure application (e.g. warnings etc.). In this case the secure SSP driver needs to combine both output buffers in a way that the secure output is always shown on top of the normal output. The logic required for this operation is called mixer logic.

To allow other applications, both on the normal and the secure side, to interact with the hardware as well, both drivers should provide facilities that expose the necessary functionality. In the secure world, the communication with the secure driver happens through the GlobalPlatform Internal Core API, version 1.1. Mission-critical control software like PLC controllers etc. can thus be separated from the hardware driver and moved to user space.

When the secure world SSP Driver wants to notify the normal world of some event, it cannot use existing APIs, as these only provide a one way communication from the normal to the secure world. This is why we provide a simple and concise package that implements just this functionality. These can be seen in the previous figure as the *controller* elements. The normal controller provides an API to register drivers to the notification system using the function `norm_ssp_register(sec_uuid, notif_handler)`. This method accepts an UUID, which is used to uniquely identify each secure application and a handler for incoming notifications. After registering the notification handler, the secure SSP driver can notify the normal SSP driver by calling `sec_ssp_notify(sec_uuid)`. This will internally trigger a hardware interrupt that will be caught by the normal controller, which in turn calls the notification handler of the normal SSP driver. The complete and detailed controlThis flow through the system can be seen in Figure 7.3.

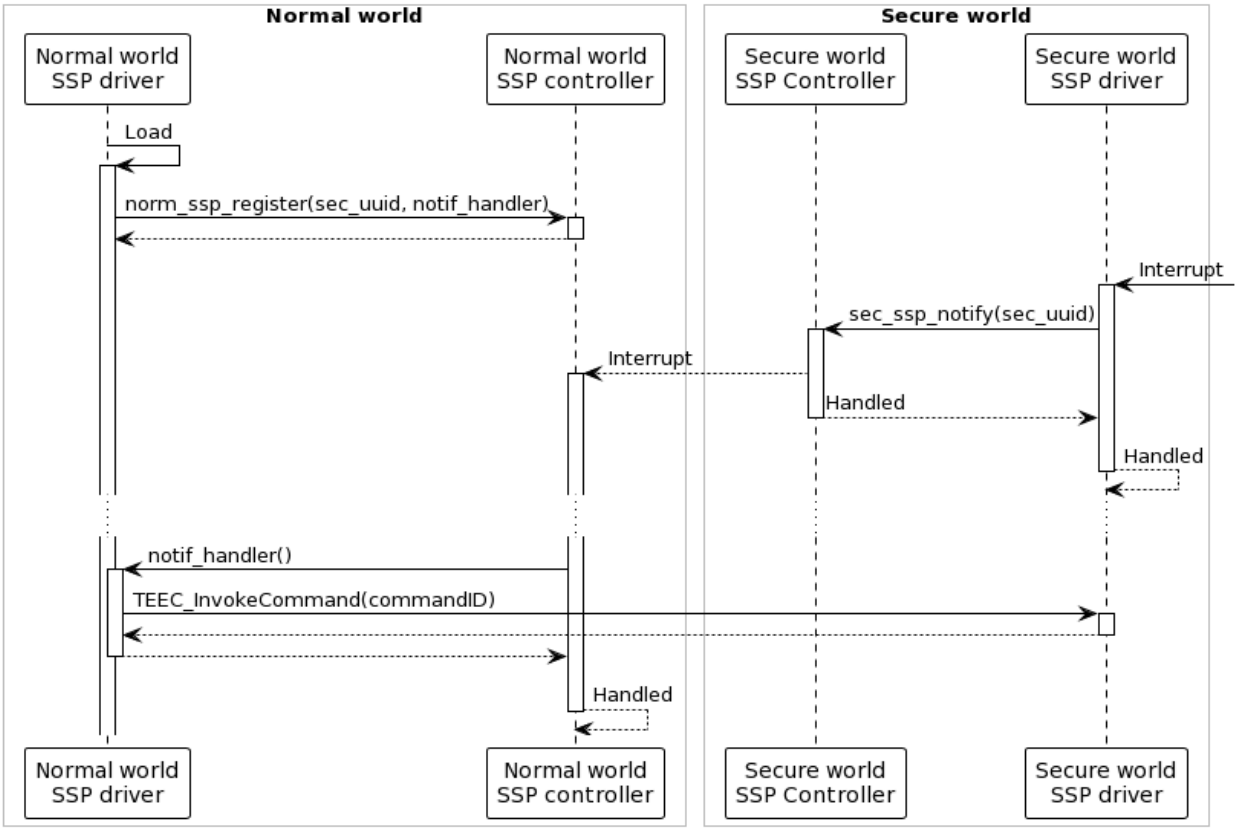


Figure 7.3: Control flow of the SSP system.

8 Linux kernel Modules (aka device drivers)

In Section 7.1 a mechanism was explained to notify the normal world of a secure world interrupt. We can now use this mechanism to implement a *Shared Secure Peripheral driver* that is responsible for relaying these secure interrupts in the secure world to a device interface in the normal world. It would then be possible to, for instance, write a normal world application that makes use of standardized device interfaces to interact with systems running in the secure world.

On Linux based operating systems, device drivers are called modules. These modules can be *built-in*, meaning they are compiled together with the kernel, or *loadable* at runtime.

8.1 Built-in modules

Built-in modules are easy to make. It suffices to include very little code into the `drivers` directory in the Linux repository. The structure of an archetypical module is as follows, and for each file a hello world example is given.

```
drivers
+- ...
+- my-module
|  +- my-module.c
|  +- Makefile
+- ...
+- Makefile
```

```
// drivers/my-module/my-module.c
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/init.h>
```

```c
#define AUTHOR  "Hello World author"
#define DESC    "Hello World driver"

static int hello_init(void)
{
   printk(KERN_ALERT "Hello, world\n");
   return 0;
}


static void hello_exit(void)
{
   printk(KERN_ALERT "Goodbye, world\n");
}


module_init(hello_init);
module_exit(hello_exit);

MODULE_LICENSE("Dual BSD/GPL");
MODULE_VERSION("v0.1");
MODULE_AUTHOR(AUTHOR);
MODULE_DESCRIPTION(DESC);
```

```makefile
# drivers/my-module/Makefile

obj-y += my-module.o
```

```makefile
# drivers/Makefile

...

obj-y += my-module/
```

8.2 Loadable kernel modules

Built-in modules are however not very interesting, as these do not allow for speedy development (the whole kernel needs to be compiled every time). Much more interesting are loadable kernel modules. These are separated pieces of code that are compiled against a kernel, making the development cycle much faster. These modules can then be very easily loaded and installed in the kernel during runtime.

It is however very easy to compile the same module code to a standalone loadable module. First, extract all code to a different development directory. Next, the Makefile needs to be changed to call the kernel build system (modules need to be compiled against the Linux kernel). If we are calling make, the environment variable `KERNELRELEASE` will not be set, and `make modules` should be called in the Linux source directory with the current working directory set as well. This will again call make in our module's directory, but now `KERNELRELEASE` will be set and the kernel build system will be used as described in the previous section.

```
# If KERNELRELEASE is defined, we've been invoked from the
# kernel build system and can use its language.
ifneq ($(KERNELRELEASE),)
   obj-m := my-module.o

# Otherwise we were called directly from the command
# line; invoke the kernel build system.
else
   KERNELDIR ?= /lib/modules/$(shell uname -r)/build # or another directory
    ↪  containing the Linux source code
   PWD := $(shell pwd)

default:
   $(MAKE) -C $(KERNELDIR) M=$(PWD) modules

endif
```

Running make will generate a module `.ko` file which can be loaded using `insmod`. `lsmod` shows all loaded modules and `rmmod` removes modules from the system.

9 Implementation into OP-TEE OS

9.1 Build instructions

9.1.1 QEMU

As shown in the figures in the previous sections, both loadable kernel modules and pseudo trusted applications are part of the architecture and thus they all need to be compiled. This requires multiple compile commands in different directories. These commands need to be executed in order:

1. Init repo with the right manifest in the project root directory:

```
mkdir -p <project-dir>
cd <project-dir>
repo init -u https://github.com/Distrinet-TACOS/manifest.git -m
 ↪  tacos-qemu-driver.xml
repo sync -j4 --no-clone-bundle
```

2. In `<project-dir>/build/`:

```
make CFG_CORE_ASLR=n GDBSERVER=y -j`nproc`
```

3. In `<project-dir>/optee_client/`:

```
make
```

4. In `<project-dir>/optee_os/`:

```
make \
    CFG_TEE_BENCHMARK=n \
    CFG_TEE_CORE_LOG_LEVEL=3 \
    CROSS_COMPILE=arm-linux-gnueabihf- \
    CROSS_COMPILE_core=arm-linux-gnueabihf- \
```

```
    CROSS_COMPILE_ta_arm32=arm-linux-gnueabihf- \
    CROSS_COMPILE_ta_arm64=aarch64-linux-gnu- \
    DEBUG=1 \
    O=out/arm \
    PLATFORM=vexpress-qemu_virt
```

5. In `<project-dir>/optee_examples/`:

```
make \
    --no-builtin-variables \
    TA_DEV_KIT_DIR=<project-dir>/optee_os/out/arm/export-ta_arm32/ \
    CROSS_COMPILE=arm-linux-gnueabihf- \
    TEEC_EXPORT=<project-dir>/optee_client/out/export/usr \
    PLATFORM=vexpress-qemu_virt \
    HOST_CROSS_COMPILE=<project-dir>/toolchains/aarch32/bin/arm-linux-
↪   gnueabihf-
```

It might be necessary to first go into specific directories in `<project-dir>/optee_examples/` because of module dependencies. The parent modules should thus be build first using the same command.

6. Finally, in `<project-dir>/build/`:

```
make run CFG_CORE_ASLR=n GDBSERVER=y -j`nproc`
```

9.1.2 SABRE Lite

As explained in Section 2.2, the *Buildroot* build system is used to build the complete package for the Sabre Lite board. For this process, the same steps are followed as explained in that section. However, an extra external Buildroot tree is added which contains the kernel modules for the normal world, while the secure world components are contained on a custom branch of our OP-TEE OS fork. It is thus easiest to start with an empty workspace.

1. Start by cloning the necessary repositories:

```
mkdir -p <project-dir>
cd <project-dir>
git clone https://github.com/Distrinet-TACOS/buildroot.git
```

```
git clone
 ↪  https://github.com/Distrinet-TACOS/buildroot-external-boundary.git
git clone
 ↪  https://github.com/Distrinet-TACOS/shared-secure-peripherals.git
```

2. Now generate the output folder using the following command:

```
make BR2_EXTERNAL=$PWD/buildroot-external-boundary/:$PWD/shared-secure-
 ↪  peripherals/ -C buildroot/ O=$PWD/output
 ↪  imx6q_sabrelite_defconfig
```

3. The build and flash steps are the same as in Section 2.2.
4. To actually test the functionality, after booting and logging in, enter the following on the normal world console:

```
modprobe normal-controller
modprobe normal-ssp-driver
```

   This loads the normal world controller and driver into the Linux kernel.

5. It is now possible to interact with the secure world serial console. Everything that you type in that console gets sent to the normal world driver, which acts as a character device. To display its buffer, execute:

```
cat /dev/normal-ssp-driver
```

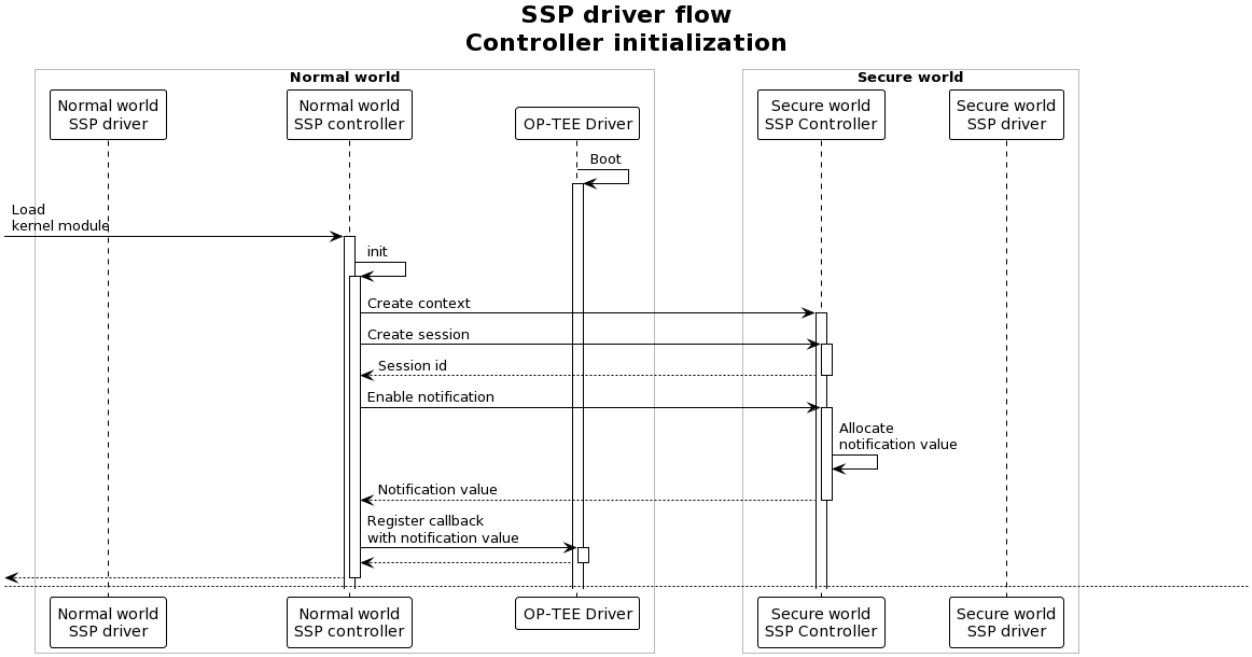
9.2 Technical details

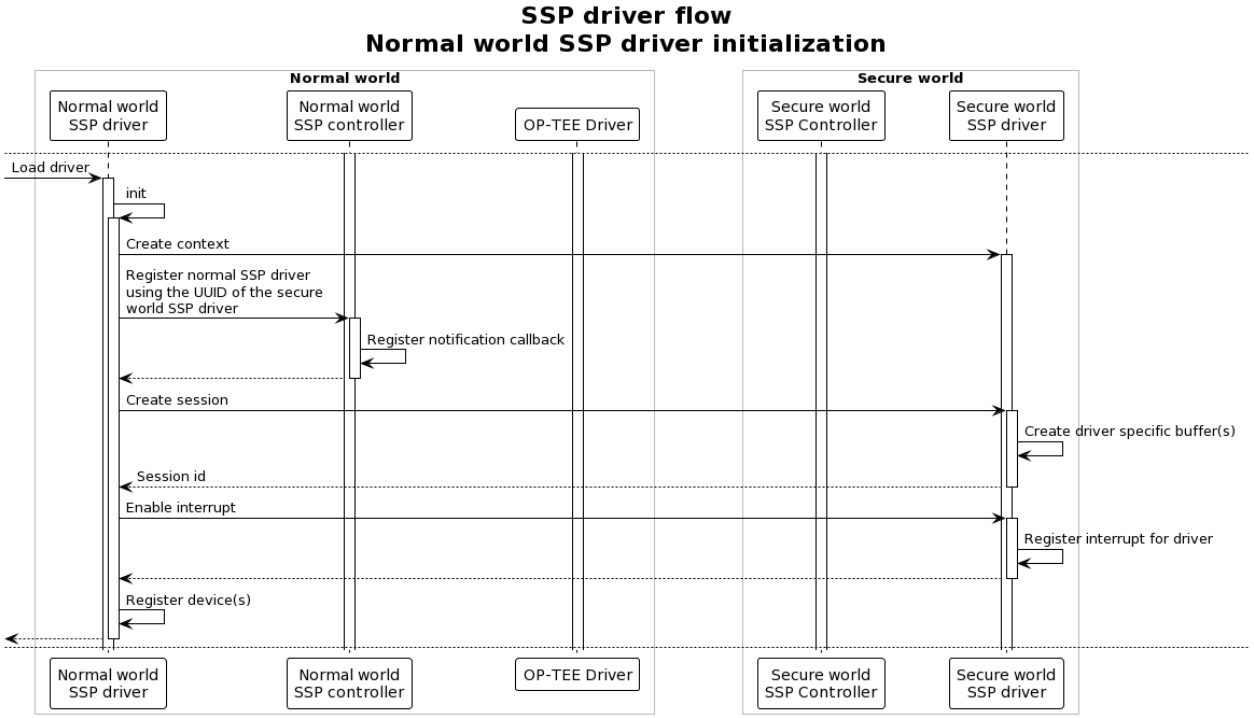
During the implementation of these shared secure peripherals on the SABRE Lite board we encountered some interesting difficulties:

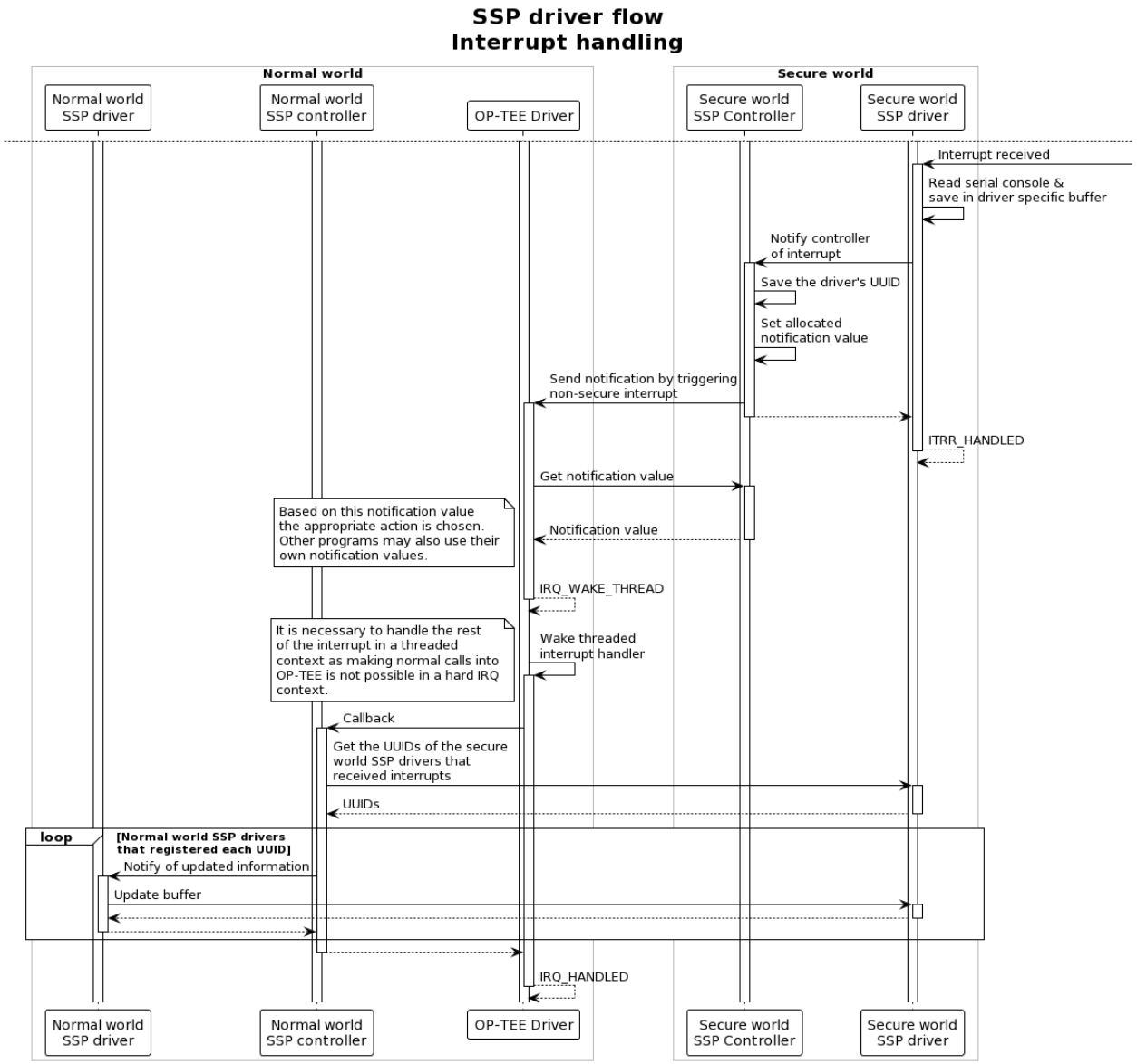
- The UART driver present for iMX.6 boards did not enable serial interrupts at boot. We added the necessary logic so that it does. The initialization of the UART subsystem is carried over from U-Boot after which the necessary bits are set.
- The device tree for the iMX.6 SABRE Lite boards did not contain an interrupt parent for the optee firmware node. The fix was to introduce the parent `intc: interrupt-controller@a01000` at the main node of the tree.

- While trying to handle the notification from the secure world in the hard IRQ context of the asynchronous notification, we had difficulties with doing a normal call into the secure world. After investigating this it became clear that this was not supported behavior. The call into the secure world is thus made in a threaded IRQ context.

The actual and detailed control flow of our implementation can be seen in the following figures:




SSP driver flow
Normal world SSP driver initialization
Normal world
Secure world
Normal world SSP driver
Normal world SSP controller
OP-TEE Driver
Secure world SSP Controller
Secure world SSP driver
Load driver
init
Create context
Register normal SSP driver using the UUID of the secure world SSP driver
Register notification callback
Create session
Create driver specific buffer(s)
Session id
Enable interrupt
Register interrupt for driver
Register device(s)
Normal world SSP driver
Normal world SSP controller
OP-TEE Driver
Secure world SSP Controller
Secure world SSP driver


SSP driver flow
Interrupt handling
Normal world
Secure world
Normal world SSP driver
Normal world SSP controller
OP-TEE Driver
Secure world SSP Controller
Secure world SSP driver
Interrupt received
Read serial console & save in driver specific buffer
Notify controller of interrupt
Save the driver's UUID
Set allocated notification value
Send notification by triggering non-secure interrupt
ITRR_HANDLED
Get notification value
Based on this notification value the appropriate action is chosen. Other programs may also use their own notification values.
Notification value
IRQ_WAKE_THREAD
It is necessary to handle the rest of the interrupt in a threaded context as making normal calls into OP-TEE is not possible in a hard IRQ context.
Wake threaded interrupt handler
Callback
Get the UUIDs of the secure world SSP drivers that received interrupts
UUIDs
loop
[Normal world SSP drivers that registered each UUID]
Notify of updated information
Update buffer
IRQ_HANDLED
Normal world SSP driver
Normal world SSP controller
OP-TEE Driver
Secure world SSP Controller
Secure world SSP driver

References

[1] "Build — OP-TEE documentation documentation." [Online]. Available: https://optee.readthedocs.io/en/latest/building/gits/build.html. [Accessed: Jan. 02, 2022]

[2] J. Corbet, A. Rubini, G. Kroah-Hartman, and A. Rubini, *Linux device drivers*, 3rd ed. Beijing ; Sebastopol, CA: O'Reilly, 2005.

[3] *OP-TEE build.git*. ForgeRock, 2021 [Online]. Available: https://github.com/ForgeRock/optee-build/blob/36ff2a361cdc3368ca17ffd4117c36b90ec63406/docs/debug.md. [Accessed: Jan. 02, 2022]

[4] "Prerequisites — OP-TEE documentation documentation." [Online]. Available: https://optee.readthedocs.io/en/latest/building/prerequisites.html. [Accessed: Jan. 02, 2022]